Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Predictive Modeling, Ensemble Methods, Classification & Scoring, Agentic AI & Workflows, RAG & LLM Orchestration, Time-Series Forecasting (SARIMA), Dimensional Modeling (Star/Snowflake, SCD Type 1/2), MLOps & Production Deployment, Automated Data Quality & Lineage, Statistical Analysis, Executive BI & Decision Intelligence

Programming Languages & Tools: Python, SQL, PySpark, AWS (S3, Redshift, EC2, DynamoDB), Azure (ADF, Databricks, Synapse, Azure SQL), Snowflake, Tableau, Power BI (DAX), Airflow, Collibra, KIRO IDE, Scikit-Learn, XGBoost, SARIMA

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Business Intelligence & AI Analytics Engineer — **May 2022 – Present**

- **Architected distributed ETL pipelines and dimensional models** using Azure, AWS Redshift, Snowflake, and PySpark to process hundreds of terabytes of engagement data, maintaining 99.9% data availability SLAs and powering core analytics.
- **Developed and deployed predictive machine learning models** including XGBoost, Random Forests, and Scikit-Learn classification ensembles to evaluate customer behavior and prioritize seller signals, boosting service reliability and decision accuracy by 20%.
- **Engineered autonomous agentic workflows and AI development tools** utilizing prompt orchestration and KIRO IDE to automate schema discovery, regex classification, and code migration, accelerating delivery turnarounds by 40-50%.
- **Designed automated data quality reconciliation firewalls and lineage** leveraging Collibra and Python scripting to validate data flows, reducing operational resolution turnaround by over 80% and ensuring pristine pipeline integrity.
- **Created executive Key Operating Metric (KOM) dashboards and semantic models** using Power BI (DAX) and Tableau, translating complex transactional telemetry and pipeline metrics into actionable executive corporate strategy and operational scorecards.

Data Scientist & Machine Learning Specialist — **May 2022 – Present**

- **Engineered autonomous machine learning anomaly detection frameworks** identifying schema variations, distribution drift, and ETL discrepancies with 95% precision to safeguard downstream BI systems and analytical reporting integrity.
- **Built advanced time-series predictive forecasting models** using SARIMA to project field support demand and operational workloads, optimizing workforce capacity allocation and reducing response times by 20%.
- **Engineered automated testing and evaluation frameworks for conversational AI** platforms like SpectrumIQ, cutting manual QA cycle times by 50% and streamlining production release cycles for enterprise AI-powered interfaces.

Business Intelligence & Decision Analytics Specialist — **May 2022 – Present**

- **Translated complex data sets into digestible business intelligence** utilizing advanced SQL, Python scripting, and Excel modeling to answer critical business questions and support operational decision-making.
- **Partnered directly with business teams and stakeholders** to identify operational improvement opportunities, define new performance metrics, and drive reporting automation across business units.
- **Mentored junior analysts on data analysis best practices** SQL query optimization, data warehousing patterns, and robust reporting methodologies to scale organizational intelligence delivery.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.